I read a lecture note which says "Traditionally, Lie brackets are defined as vector fields. Indeed, the tangent space $\Lie\bbG=T_e\bbG$ can be identified via left translation to $T_g\bbG$ to any $g\in\bbG(\bar{F})$. Hence $\Lie\bbG$ can be identified with the space of left-invariant algebraic vector fields on $\bbG$. Likewise $\fg$ can be identified with the space of left-invariant analytic vector fields on G. In both the algebraic and analytic case, the Lie bracket agree with the commutator in the sense of derivations."

In this lecture note, the ground field is a p-adic field so that the "analytic" geometry is for p-adic analytic manifolds. Can you explain the Lie bracket and derivation part to me?

The key idea is that **vector fields act as derivations on functions**, and the **Lie bracket of vector fields is exactly the commutator of these derivations**. The sentence in your lecture note is referring to this identification.

Let me unpack the pieces carefully.

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1. Vector fields as derivations

Let M be a manifold (or algebraic variety).
A **vector field** X can be viewed in two equivalent ways:

1. **Geometric definition**
   A section of the tangent bundle

$$X \in \Gamma(M, TM).$$

2. **Algebraic definition**
   A **derivation** of the algebra of functions.

If f is a function on M, the vector field differentiates it:

$$X(f)$$

This operation satisfies the **Leibniz rule**

$$X(fg) = X(f)g + fX(g).$$

So

$$X : C^\infty(M) \to C^\infty(M)$$

is a **derivation of the algebra of functions**.

Algebraic geometry version

If X is an algebraic variety, a vector field is

$$X \in \mathrm{Der}_k(\mathcal{O}_X, \mathcal{O}_X)$$

a derivation of the structure sheaf.

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p-adic analytic version

For a p-adic analytic manifold G,

$$X : C^{an}(G) \to C^{an}(G)$$

is a derivation of analytic functions.

So the notion of vector field = derivation works **in all three settings**:

- smooth manifolds
- algebraic varieties
- p-adic analytic manifolds

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2. Commutator of derivations

If X and Y are derivations, we can compose them:

$$X(Y(f)), \quad Y(X(f)).$$

The **commutator** is

$$[X, Y] := X \circ Y - Y \circ X.$$

This is again a derivation.

Indeed,

$$[X,Y](fg) = [X,Y](f)g + f[X,Y](g).$$

So derivations form a **Lie algebra** under this commutator.

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3. Lie bracket of vector fields

Traditionally the **Lie bracket of vector fields** is defined geometrically.

If

$$X = \sum a_i(x)\frac{\partial}{\partial x_i}, \quad Y = \sum b_i(x)\frac{\partial}{\partial x_i},$$

then

$$[X,Y] = \sum_i \left(X(b_i) - Y(a_i)\right)\frac{\partial}{\partial x_i}.$$

One checks that

$$[X,Y](f) = X(Y(f)) - Y(X(f)).$$

So the Lie bracket of vector fields **agrees exactly with the commutator of the corresponding derivations**.

This is what your lecture note means.

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4. Application to Lie groups

Now take a group G.

A vector $X \in T_eG$ determines a **left-invariant vector field**

$$X_g = d(L_g)_e(X)$$

where $L_g(h) = gh$.

Thus

$$\Lie G \cong \{\text{left-invariant vector fields on } G\}.$$

Now take two elements $X, Y \in \Lie G$.

View them as left-invariant vector fields.

Compute their bracket as vector fields:

$$[X, Y].$$

One checks:

- it is again left-invariant
- its value at e gives an element of T_eG

So

$$[X, Y]_{\Lie G} = ([X, Y]_{\text{vector fields}})(e).$$

But since vector fields act as derivations, this is

$$[X, Y]f = X(Y(f)) - Y(X(f)).$$

Thus the Lie algebra bracket is literally **the commutator of derivations acting on functions on G**.

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5. Why the lecture note emphasizes this

They want to say:

- In the **algebraic setting**

$$\Lie\bbG = \text{left-invariant algebraic vector fields}$$

acting as derivations on $k[\bbG]$.

- In the **p-adic analytic setting**

$$\fg = \text{left-invariant analytic vector fields}$$

acting as derivations on analytic functions on G.

In both cases the bracket is

$$[X, Y] = X \circ Y - Y \circ X.$$

So the constructions are **completely parallel**.

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6. Intuition

Think of vector fields as **differential operators**.

Then

- X = differentiate in direction X
- Y = differentiate in direction Y

That failure is exactly

$$X(Y(f)) - Y(X(f)).$$

✅ Summary

The sentence in the lecture note means:

- Vector fields act as derivations on functions.
- Derivations form a Lie algebra via the commutator.
- The Lie bracket of vector fields equals this commutator.
- Therefore the Lie algebra of a group can be realized as left-invariant vector fields with bracket

$$[X, Y] = X \circ Y - Y \circ X.$$

If you'd like, I can also show you the **very concrete computation for a matrix group** (e.g. GL_n) where this derivation description directly produces the familiar bracket $XY - YX$. That example makes the statement in the note extremely transparent.